

COMPLETE PHYSICS

Scientific Stream / Upper Secondary / Terminale

Concepts • Formulae • Worked Examples

30 Complete Lessons • Practice Activities

Revision Tests • Answer Key • Reference Tables

Mechanics • Waves • Thermal Physics • Electricity • Modern Physics

A structured 80-page learning book

Designed for classroom teaching, independent study, scientific problem solving, and exam preparation. Each lesson combines theory, formulae, worked examples, common errors, guided practice, and mastery checks.

PHYSICS SCIENTIFIC SERIES

GETTING STARTED

How to Use This Book

A practical roadmap for mastering physics step by step.

What You Will Learn

- Core upper-secondary physics from measurement and mechanics to electricity, optics, atomic and nuclear physics.
- A disciplined problem-solving method: diagram -> known values -> principle -> algebra -> units -> reasonableness check.
- How to interpret graphs, use equations, explain models, and identify assumptions and limitations.
- Exam-focused practice through worked examples, common-error correction, revision tests, and answer keys.

Recommended Study Method

1. Read the learning objectives and key concepts.
2. Copy the formulae and state the meaning and SI unit of each symbol.
3. Study the worked example, then cover it and solve it again.
4. Complete guided practice without checking answers.
5. Record every mistake with the corrected principle or equation.
6. Revisit difficult lessons after 24 hours and after one week.

Physics Problem-Solving Checklist

- Diagram: Is the situation represented clearly?
- Known/unknown: Are quantities written with SI units?
- Principle: Which law or model applies, and under what assumptions?
- Algebra: Can the equation be rearranged symbolically before substitution?
- Units: Do both sides of the equation have compatible dimensions?
- Reasonableness: Is the sign, direction, and magnitude physically sensible?

Suggested 6-Week Study Schedule

Week 1: Lessons 1-5 - measurement, vectors, kinematics, projectiles, Newton laws.
 Week 2: Lessons 6-10 - circular motion, gravitation, energy, momentum, rotation.
 Week 3: Lessons 11-15 - oscillations, waves, sound, fluids, thermal physics.
 Week 4: Lessons 16-20 - gases, thermodynamics, electrostatics, potential, DC circuits.
 Week 5: Lessons 21-25 - Kirchhoff, magnetism, induction, AC, geometrical optics.
 Week 6: Lessons 26-30 + all revision tests - wave optics, EM, atomic, nuclear, semiconductors.

REFERENCE

Table of Contents

Lessons 1-15

Lesson	Topic	Pages
1	Measurement, Units and Uncertainty	5-6
2	Scalars and Vectors	7-8
3	Motion in One Dimension	9-10
4	Projectile Motion	11-12
5	Newton's Laws of Motion	13-14
6	Friction and Circular Motion	15-16
7	Gravitation	17-18
8	Work, Energy and Power	19-20
9	Momentum and Collisions	21-22
10	Rotational Motion and Torque	23-24
11	Simple Harmonic Motion	25-26
12	Mechanical Waves	27-28
13	Sound	29-30
14	Fluid Statics and Dynamics	31-32
15	Thermal Physics and Calorimetry	33-34

Learning Map

Lessons 1-5

Foundations: measurement, vectors, kinematics, projectiles, Newton laws.

Lessons 6-10

Forces and conservation: friction, gravitation, energy, momentum, rotation.

Lessons 11-15

Oscillations, waves, sound, fluids, and thermal physics.

Milestones

- I can select an appropriate physical model and state its assumptions.
- I can solve numerical problems with units, signs, and significant figures.
- I can interpret graphs, explain trends, and connect equations to physical meaning.
- I can check answers using dimensions, limiting cases, and order-of-magnitude reasoning.

REFERENCE

Table of Contents

Lessons 16-30, Revision and Reference

Lesson	Topic	Pages
16	Kinetic Theory and the Ideal Gas	35-36
17	Laws of Thermodynamics	37-38
18	Electrostatics	39-40
19	Electric Potential and Capacitance	41-42
20	Direct-Current Circuits	43-44
21	Kirchhoff's Laws and Electrical Power	45-46
22	Magnetic Fields and Forces	47-48
23	Electromagnetic Induction	49-50
24	Alternating Current and Transformers	51-52
25	Geometrical Optics	53-54
26	Wave Optics	55-56
27	Electromagnetic Waves	57-58
28	Atomic Physics	59-60
29	Nuclear Physics	61-62
30	Semiconductors and Modern Physics Review	63-64

Learning Map

Lessons 16-20	Gases, thermodynamics, electrostatics, potential/capacitance, circuits.
Lessons 21-25	Network circuits, magnetism, induction, AC, and geometrical optics.
Lessons 26-30	Wave optics, EM waves, atomic, nuclear, and semiconductor physics.

Milestones

- I can select an appropriate physical model and state its assumptions.
- I can solve numerical problems with units, signs, and significant figures.
- I can interpret graphs, explain trends, and connect equations to physical meaning.
- I can check answers using dimensions, limiting cases, and order-of-magnitude reasoning.

Revision and Reference

Section	Topic	Pages
Revision	Comprehensive Review Tests	65-72
Answers	Answer Key	73-78
Reference	Formulae, Constants and SI Units	79
Reference	Physics Quick Guide	80

LESSON 01 - CORE PHYSICS

Measurement, Units and Uncertainty

Concepts, formulae, key rules, and worked example

Learning Objectives

- Use SI base and derived units correctly.
- Apply scientific notation, significant figures, and dimensional analysis.
- Distinguish random uncertainty from systematic error.

Concept	Explanation	Formula / Example
SI system	Physics measurements are expressed with agreed base units and derived units.	length: m; mass: kg; time: s
Scientific notation	Very large or small values are written as $a \times 10^n$.	$0.00032 \text{ m} = 3.2 \times 10^{-4} \text{ m}$
Significant figures	Report digits justified by the precision of the measurement.	12.30 cm has 4 significant figures
Uncertainty	A measured value is stated with an estimated range.	$L = (25.4 \pm 0.1) \text{ cm}$

Key Rules

- Convert all quantities to compatible units before substituting into a formula.
- A final numerical answer should normally reflect the least precise measured input.
- Dimensions can test whether an equation is physically plausible, but not whether its numerical constant is correct.

Worked Example

A student measures 2.40 m in 1.20 s. Average speed = distance/time = $2.40/1.20 = 2.00 \text{ m/s}$. Both measurements have 3 significant figures, so report 2.00 m/s.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 01 - PRACTICE

Measurement, Units and Uncertainty

Common errors, guided practice, and mastery check

Common error	Better form	Why?
5 km = 5 m	5 km = 5000 m	The prefix kilo means 10^3 .
$v = 3.456789 \text{ m/s}$ from 2 s timing	Round to justified precision	Calculated digits cannot create measurement precision.

Guided Practice

- Convert 72 km/h to m/s.
- Write 0.0000450 m in scientific notation.
- How many significant figures are in 0.02030?
- State the SI unit of force.
- Find the percentage uncertainty in $5.0 \pm 0.2 \text{ cm}$.
- Check the dimensions of $v = u + at$.
- Distinguish accuracy from precision.
- Why should repeated measurements be averaged?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- 20 m/s
- $4.50 \times 10^{-5} \text{ m}$
- 4
- newton ($\text{N} = \text{kg m/s}^2$)
- 4%
- Each term has dimensions L/T.
- Accuracy = closeness to accepted value; precision = repeatability.
- To reduce the effect of random variation.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Scalars and Vectors

Concepts, formulae, key rules, and worked example

Learning Objectives

- Distinguish scalar quantities from vector quantities.
- Resolve vectors into perpendicular components.
- Add vectors graphically and analytically.

Concept	Explanation	Formula / Example
Scalar	Has magnitude only.	mass, time, energy, temperature
Vector	Has magnitude and direction.	displacement, velocity, force
Components	A vector can be resolved along chosen axes.	$A_x = A \cos(\theta)$; $A_y = A \sin(\theta)$
Resultant	Vector sum found by components or geometry.	$R = A + B$

Key Rules

- Choose a positive coordinate direction before assigning signs.
- Perpendicular components recombine with Pythagoras: $R = \sqrt{R_x^2 + R_y^2}$.
- Direction must be stated with an angle and a reference axis or compass direction.

Worked Example

A 10 N force acts 30 degrees above the horizontal. $F_x = 10 \cos 30 = 8.66 \text{ N}$ and $F_y = 10 \sin 30 = 5.00 \text{ N}$.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

Scalars and Vectors

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Speed is a vector	Velocity is a vector	Speed has magnitude only.
Add 3 N east + 4 N north = 7 N	Resultant = 5 N northeast	Perpendicular vectors add geometrically, not arithmetically.

Guided Practice

1. Classify distance: scalar or vector?

2. Classify acceleration.

3. Resolve 20 N at 60 degrees into x and y components.

4. Find the resultant of 6 N east and 8 N north.

5. Find the direction of that resultant north of east.

6. What is the resultant of equal opposite vectors?

7. Why is displacement different from distance?

8. A vector has components 3 and 4. Find its magnitude.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. scalar

2. vector

3. 10 N, 17.3 N

4. 10 N

5. 53.1 degrees

6. zero

7. Displacement includes direction and depends only on start/end points.

8. 5

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 03 - CORE PHYSICS

Motion in One Dimension

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe motion using displacement, velocity, and acceleration.
- Interpret displacement-time and velocity-time graphs.
- Apply constant-acceleration equations.

Concept	Explanation	Formula / Example
Displacement	Change in position with direction.	$\Delta x = x_f - x_i$
Velocity	Rate of change of displacement.	$v = \Delta x / \Delta t$
Acceleration	Rate of change of velocity.	$a = \Delta v / \Delta t$
SUVAT equations	Used when acceleration is constant.	$v = u + at$; $s = ut + 0.5at^2$

Key Rules

- Do not use constant-acceleration equations when acceleration varies significantly.
- Slope of x-t graph gives velocity; slope of v-t graph gives acceleration.
- Area under a v-t graph gives displacement.

Worked Example

A car starts from rest and accelerates at 3.0 m/s^2 for 4.0 s . $v = 0 + 3.0(4.0) = 12 \text{ m/s}$. $s = 0.5(3.0)(4.0^2) = 24 \text{ m}$.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

LESSON 03 - PRACTICE

Motion in One Dimension

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Use speed in every signed equation	Use signed velocity	Direction is represented by the sign.
Area under x-t graph gives distance	Slope of x-t gives velocity	Graph meanings depend on the axes.

Guided Practice

- Define instantaneous velocity.
- A body moves from $x=2$ m to $x=-3$ m. Find displacement.
- From rest, $a=2$ m/s² for 5 s. Find v .
- For the same motion, find s .
- What does a horizontal v-t line mean?
- What does negative acceleration mean?
- A car slows from 20 to 5 m/s in 3 s. Find a .
- State one condition for using $v^2 = u^2 + 2as$.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- Velocity at a particular instant.
- 5 m
- 10 m/s
- 25 m
- constant velocity
- Acceleration points in the negative chosen direction.
- 5 m/s²
- Acceleration is constant.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 04 - CORE PHYSICS

Projectile Motion

Concepts, formulae, key rules, and worked example

Learning Objectives

- Resolve projectile motion into horizontal and vertical components.
- Calculate time of flight, maximum height, and range in ideal cases.
- Explain why horizontal and vertical motions are independent.

Concept	Explanation	Formula / Example
Horizontal motion	With air resistance neglected, horizontal acceleration is zero.	$v_x = \text{constant}$
Vertical motion	Vertical acceleration is approximately $-g$.	$v_y = u_y - gt$
Launch components	Resolve launch speed into x and y parts.	$u_x = u \cos(\theta)$; $u_y = u \sin(\theta)$
Trajectory	Combined independent motions produce a parabola.	$x = u_x t$

Key Rules

- Use the same time variable for horizontal and vertical motion.
- At maximum height, vertical velocity is zero, but horizontal velocity is not.
- For launch and landing at the same level, time up equals time down in the ideal model.

Worked Example

A ball is launched at 20 m/s at 30 degrees. $u_x = 17.3$ m/s and $u_y = 10.0$ m/s. Time to highest point = $u_y/g = 10/9.8 = 1.02$ s.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 04 - PRACTICE

Projectile Motion

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Horizontal acceleration = g	Vertical acceleration = g downward	Gravity acts vertically.
Velocity is zero at the top	Only vertical component is zero	Horizontal motion continues.

Guided Practice

1. Resolve 30 m/s at 45 degrees into components.
2. What is vertical velocity at maximum height?
3. What is horizontal acceleration ideally?
4. Find time to top if $u_y=19.6$ m/s.

5. Find maximum height for $u_y=19.6$ m/s.
6. Why does a projectile curve downward?
7. State the shape of the ideal trajectory.
8. Name one real effect neglected in the simple model.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. 21.2 m/s, 21.2 m/s
2. 0 m/s
3. 0 m/s^2
4. 2.0 s

5. 19.6 m
6. Gravity changes the vertical velocity.
7. parabola
8. air resistance/wind/spin

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 05 - CORE PHYSICS

Newton's Laws of Motion

Concepts, formulae, key rules, and worked example

Learning Objectives

- State and apply Newton's three laws.
- Draw free-body diagrams for interacting objects.
- Relate net force, mass, and acceleration.

Concept	Explanation	Formula / Example
First law	If net force is zero, velocity remains constant.	$\Sigma F = 0 \rightarrow a = 0$
Second law	Net force equals rate of change of momentum; for constant mass, $F=ma$.	$\Sigma F = ma$
Third law	Interaction forces occur in equal and opposite pairs on different bodies.	$F_{AB} = -F_{BA}$
Weight	Gravitational force on a mass near Earth.	$W = mg$

Key Rules

- A free-body diagram contains only forces acting on the chosen object.
- Action-reaction forces never cancel on one free-body diagram because they act on different objects.
- Use the net force in $F=ma$, not a single force unless it is the only force.

Worked Example

A 5 kg box is pulled right by 30 N while friction is 10 N left. Net force = 20 N right, so $a = 20/5 = 4 \text{ m/s}^2$ right.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 05 - PRACTICE

Newton's Laws of Motion

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Motion requires a net force	Acceleration requires a net force	Constant velocity can occur with zero net force.
Weight = mass	Weight = mg	Mass is measured in kg; weight is force in N.

Guided Practice

- State Newton's first law.
- Find weight of 10 kg if $g=9.8 \text{ m/s}^2$.
- A 4 kg body has net force 12 N. Find a .
- Name the force exerted by a surface perpendicular to itself.
- Why do third-law forces not cancel each other on one body?
- A lift moves upward at constant speed. What is net force?
- What happens to acceleration if same force acts on double mass?
- Draw verbally the forces on a resting book on a table.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- Constant velocity unless acted on by a nonzero net force.
- 98 N
- 3 m/s^2
- normal force
- They act on different bodies.
- zero
- It halves.
- Weight down and normal force up.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Friction and Circular Motion

Concepts, formulae, key rules, and worked example

Learning Objectives

- Distinguish static and kinetic friction.
- Calculate centripetal acceleration and force.
- Identify the real force that provides centripetal force in examples.

Concept	Explanation	Formula / Example
Static friction	Adjusts up to a limiting value before sliding begins.	$f_s \leq \mu_s N$
Kinetic friction	Acts during sliding and is often approximated by $\mu_k N$.	$f_k = \mu_k N$
Centripetal acceleration	Acceleration toward the center of circular motion.	$a_c = v^2/r$
Centripetal force	Name for the net inward force, not a new interaction.	$F_c = mv^2/r$

Key Rules

- Friction opposes relative slipping or the tendency to slip, not necessarily the direction of motion.
- For uniform circular motion, speed can be constant while velocity changes.
- Always identify the actual force (friction, tension, gravity, normal) providing the inward resultant.

Worked Example

A 1000 kg car rounds a flat curve of radius 50 m at 10 m/s. Required inward force = $mv^2/r = 1000(100)/50 = 2000$ N, supplied by friction.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

LESSON 06 - PRACTICE

Friction and Circular Motion

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Centripetal force is an extra force	It is the net inward force	Use actual interaction forces in the free-body diagram.
Constant speed means no acceleration	Circular motion has inward acceleration	Velocity direction changes continuously.

Guided Practice

1. Write the formula for centripetal acceleration.	5. What force can provide it for a car on a flat road?
2. A 2 kg mass moves at 4 m/s in radius 2 m. Find F_c .	6. State one difference between static and kinetic friction.
3. Direction of centripetal acceleration?	7. If speed doubles at fixed r , how does F_c change?
4. What force can provide centripetal force for a satellite?	8. If radius doubles at fixed v , how does a_c change?

Mastery Challenge

- Solve two original numerical examples using the lesson principles. - For each, write the governing equation before inserting numbers. - Check units, significant figures, sign/direction, and physical reasonableness.
--

Quick Answers

1. v^2/r	5. friction
2. 16 N	6. Static acts before slipping; kinetic acts during sliding.
3. toward the center	7. It becomes 4 times larger.
4. gravity	8. It halves.

Exam Strategy

- Write known quantities with units before substituting numbers. - Keep extra digits during working, then round the final answer appropriately. - If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Gravitation

Concepts, formulae, key rules, and worked example

Learning Objectives

- Apply Newton's law of universal gravitation.
- Relate gravitational field strength to force per unit mass.
- Describe circular orbital motion and escape concepts.

Concept	Explanation	Formula / Example
Universal gravitation	Any two masses attract each other.	$F = Gm_1m_2/r^2$
Field strength	Force per unit mass at a point.	$g = GM/r^2$
Potential energy	Gravitational potential energy for isolated masses is negative.	$U = -GMm/r$
Circular orbit	Gravity supplies centripetal force.	$GMm/r^2 = mv^2/r$

Key Rules

- Distance r is measured center-to-center for spherical bodies.
- Gravitational field strength decreases with the square of distance from the center.
- Orbiting objects are continuously falling around the central body.

Worked Example

For a circular orbit, $GMm/r^2 = mv^2/r$. Cancel m to obtain $v = \sqrt{GM/r}$, showing orbital speed does not depend on satellite mass.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 07 - PRACTICE

Gravitation

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Use height above surface as r	Use distance from center	Inverse-square law uses separation of centers.
Astronauts in orbit have no gravity	They are in free fall	Gravity is what keeps them in orbit.

Guided Practice

1. State Newton's law of gravitation.	5. Why is orbital speed independent of satellite mass?
2. How does force change if distance doubles?	6. What provides centripetal force in orbit?
3. Define gravitational field strength.	7. Is gravitational potential energy of a bound two-mass system positive or negative?
4. Write g around a spherical mass M .	8. What happens to g far from a planet?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. $F = Gm_1m_2/r^2$	5. Satellite mass cancels from the force equation.
2. It becomes one quarter.	6. gravity
3. Force per unit mass.	7. negative
4. GM/r^2	8. It approaches zero.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 08 - CORE PHYSICS

Work, Energy and Power

Concepts, formulae, key rules, and worked example

Learning Objectives

- Calculate work by constant forces.
- Use kinetic and potential energy relations.
- Apply conservation of mechanical energy and power.

Concept	Explanation	Formula / Example
Work	Energy transferred by a force through displacement.	$W = Fd \cos(\theta)$
Kinetic energy	Energy associated with motion.	$K = 0.5mv^2$
Gravitational PE	Near Earth, change in gravitational PE.	$\Delta U = mg \Delta h$
Power	Rate of doing work or transferring energy.	$P = W/t$; $P = Fv$ for parallel motion

Key Rules

- Only the component of force along displacement does work.
- Net work equals change in kinetic energy.
- Mechanical energy is conserved when only conservative forces do work.

Worked Example

A 2 kg object falls 5 m from rest with negligible air resistance. Lost PE = $mgh = 2(9.8)(5) = 98$ J, so final kinetic energy is 98 J.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 08 - PRACTICE

Work, Energy and Power

Common errors, guided practice, and mastery check

Common error	Better form	Why?
A force always does work	Work requires displacement component along force	Perpendicular force does zero work.
Power and energy are the same	Power = energy/time	Their units are J and W respectively.

Guided Practice

- Find work by 10 N over 3 m in same direction.
- Find kinetic energy of 4 kg at 5 m/s.
- Find PE gain of 2 kg raised 3 m.
- Define power.
- A 600 J job takes 3 s. Find power.
- State the work-energy theorem.
- Does centripetal force do work in uniform circular motion?
- Why can friction reduce mechanical energy?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- 30 J
- 50 J
- 58.8 J
- Rate of energy transfer/work.
- 200 W
- Net work equals change in kinetic energy.
- No, because it is perpendicular to velocity.
- It converts mechanical energy to internal/thermal energy.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 09 - CORE PHYSICS

Momentum and Collisions

Concepts, formulae, key rules, and worked example

Learning Objectives

- Calculate linear momentum and impulse.
- Apply conservation of momentum to isolated systems.
- Distinguish elastic and inelastic collisions.

Concept	Explanation	Formula / Example
Momentum	Vector quantity equal to mass times velocity.	$p = mv$
Impulse	Change in momentum caused by force over time.	$J = \Delta p = F_{\text{avg}} \Delta t$
Conservation	Total momentum remains constant for an isolated system.	$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$
Collision type	Elastic conserves kinetic energy; inelastic does not.	Perfectly inelastic: objects stick

Key Rules

- Momentum conservation applies to a system when external impulse is negligible.
- Use signs or vector components consistently.
- Kinetic energy is not generally conserved in inelastic collisions, but total energy is.

Worked Example

A 2 kg cart moving 3 m/s hits a 1 kg cart at rest and they stick. Initial momentum = 6 kg m/s. Final speed = $6/(3) = 2$ m/s.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

LESSON 09 - PRACTICE

Momentum and Collisions

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Momentum is scalar	Momentum is vector	Its direction is the direction of velocity.
All collisions conserve kinetic energy	Only elastic collisions do	Momentum is the general collision conservation law for isolated systems.

Guided Practice

- Find momentum of 3 kg at 4 m/s.
- A 10 N force acts 0.2 s. Find impulse.
- What condition is needed for momentum conservation?
- Define elastic collision.
- Two objects stick after collision: what type?
- A 1 kg object at 5 m/s stops. Find Δp .
- How do airbags reduce force?
- Is total energy conserved in an inelastic collision?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- 12 kg m/s
- 2 N s
- Negligible external impulse.
- Momentum and kinetic energy are conserved.
- perfectly inelastic
- 5 kg m/s
- They increase stopping time for the same momentum change.
- Yes, though mechanical/kinetic energy changes form.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Rotational Motion and Torque

Concepts, formulae, key rules, and worked example

Learning Objectives

- Relate angular and linear quantities.
- Calculate torque and rotational equilibrium.
- Use rotational kinetic energy and angular momentum ideas.

Concept	Explanation	Formula / Example
Angular motion	Angular displacement, velocity, and acceleration describe rotation.	$v = r \omega$
Torque	Turning effect of a force about an axis.	$\tau = rF \sin(\theta)$
Moment of inertia	Rotational analogue of mass.	I depends on mass distribution
Rotational dynamics	Net torque produces angular acceleration.	$\Sigma \tau = I \alpha$

Key Rules

- Use the perpendicular lever arm to calculate torque.
- For static equilibrium, both net force and net torque are zero.
- Moving mass farther from the axis generally increases moment of inertia.

Worked Example

A 20 N force acts perpendicular to a 0.50 m wrench. Torque = $rF = 0.50(20)=10$ N m.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 10 - PRACTICE

Rotational Motion and Torque

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Torque = force only	Torque depends on force and lever arm	Same force can have different turning effects.
Zero net force guarantees equilibrium	Need zero net torque too	An object can rotate with balanced forces.

Guided Practice

1. Define torque.	5. State two conditions for static equilibrium.
2. Find torque of 15 N perpendicular at 0.4 m.	6. What happens to I if mass moves farther from axis?
3. What is the SI unit of torque?	7. Write rotational Newton second law.
4. Relate linear speed and angular speed.	8. Why is a long wrench useful?

Mastery Challenge

- Solve two original numerical examples using the lesson principles. - For each, write the governing equation before inserting numbers. - Check units, significant figures, sign/direction, and physical reasonableness.
--

Quick Answers

1. Turning effect of a force about an axis.	5. $\Sigma F=0$ and $\Sigma \tau=0$
2. 6 N m	6. It increases.
3. N m	7. $\Sigma \tau = I \alpha$
4. $v = r \omega$	8. It increases lever arm and torque for the same force.

Exam Strategy

- Write known quantities with units before substituting numbers. - Keep extra digits during working, then round the final answer appropriately. - If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Simple Harmonic Motion

Concepts, formulae, key rules, and worked example

Learning Objectives

- Recognize the defining condition for simple harmonic motion.
- Relate displacement, velocity, acceleration, period, and energy.
- Apply SHM models to springs and pendulums.

Concept	Explanation	Formula / Example
SHM condition	Acceleration is proportional to displacement and opposite in direction.	$a = -\omega^2 x$
Spring oscillator	Hooke restoring force can produce SHM.	$F = -kx$
Period	Spring period depends on m and k.	$T = 2\pi \sqrt{m/k}$
Energy	Total mechanical energy is constant without damping.	$E = 0.5kA^2$

Key Rules

- At equilibrium, speed is maximum and acceleration is zero.
- At extreme displacement, speed is zero and acceleration magnitude is maximum.
- The simple pendulum period formula applies for small angles.

Worked Example

For $m=0.50\text{ kg}$ and $k=200\text{ N/m}$, $T = 2\pi \sqrt{0.50/200} = 0.314\text{ s}$.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 11 - PRACTICE

Simple Harmonic Motion

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Acceleration is zero at extremes	Acceleration is maximum at extremes	Restoring force is largest there.
SHM speed is constant	Speed varies periodically	Energy moves between kinetic and potential forms.

Guided Practice

- State the SHM acceleration relation.
- Where is speed maximum?
- Where is acceleration magnitude maximum?
- Write spring period formula.
- Find T for $m=1$ kg, $k=100$ N/m.
- What is amplitude?
- How does damping affect amplitude?
- For a small pendulum, what variables determine period?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $a = -\omega^2 x$
- equilibrium
- at maximum displacement
- $2\pi \sqrt{m/k}$
- about 0.628 s
- Maximum displacement from equilibrium.
- It decreases with time.
- length and g (not mass)

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Mechanical Waves

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe waves using amplitude, wavelength, frequency, and speed.
- Distinguish transverse and longitudinal waves.
- Apply superposition, reflection, and standing-wave ideas.

Concept	Explanation	Formula / Example
Wave speed	Speed equals frequency times wavelength.	$v = f \lambda$
Transverse	Oscillation is perpendicular to wave travel.	wave on a string
Longitudinal	Oscillation is parallel to wave travel.	sound in air
Standing waves	Produced by interference of opposite traveling waves.	nodes and antinodes

Key Rules

- Frequency is set by the source; wavelength changes if wave speed changes.
- Waves transfer energy without permanent bulk transfer of matter.
- At a fixed-end reflection, a transverse pulse inverts.

Worked Example

A wave has $f=5\text{ Hz}$ and $\lambda=2.4\text{ m}$. Its speed is $v=f \lambda=12\text{ m/s}$.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

Mechanical Waves

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Amplitude determines wave speed in a linear medium	Medium properties determine speed	Amplitude mainly affects energy.
Frequency changes at a boundary	Frequency remains the same	Speed and wavelength may change instead.

Guided Practice

1. Write the wave equation.

2. Find speed for $f=20\text{ Hz}$, $\lambda=0.50\text{ m}$.

3. Define wavelength.

4. Give one transverse example.

5. Give one longitudinal example.

6. What is a node?

7. What happens to frequency when entering a new medium?

8. What principle explains interference?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. $v = f \lambda$

2. 10 m/s

3. Distance between points in the same phase on successive cycles.

4. string/electromagnetic wave

5. sound in air

6. Point of zero displacement in a standing wave.

7. It remains constant.

8. superposition

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 13 - CORE PHYSICS

Sound

Concepts, formulae, key rules, and worked example

Learning Objectives

- Explain sound as a longitudinal mechanical wave.
- Relate frequency to pitch and intensity to loudness.
- Apply resonance, beats, and Doppler-effect concepts.

Concept	Explanation	Formula / Example
Sound wave	Pressure/density variations traveling through a medium.	requires material medium
Intensity	Power transmitted per unit area.	$I = P/A$
Resonance	Large response when driving frequency matches natural frequency.	$f_{\text{drive}} \sim f_{\text{natural}}$
Doppler effect	Observed frequency changes with relative source-observer motion.	approaching -> higher observed frequency

Key Rules

- Sound cannot travel through a perfect vacuum.
- Intensity follows an inverse-square trend for a point source in open space.
- Pitch is primarily associated with frequency, not amplitude.

Worked Example

If a point source spreads power uniformly, doubling distance makes area four times larger, so intensity becomes one quarter.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 13 - PRACTICE

Sound

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Louder means higher frequency	Louder mainly means greater intensity	Frequency mainly changes pitch.
Sound travels in vacuum	Sound needs a medium	It is a mechanical wave.

Guided Practice

- Is sound transverse or longitudinal in air?
- What determines pitch?
- What determines intensity physically?
- How does intensity change when distance doubles from a point source?
- Define resonance.
- What are beats?
- Approaching source: observed frequency higher or lower?
- Why is there no sound on the Moon without radio equipment?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- longitudinal
- frequency
- power per unit area / amplitude-related energy
- one quarter
- Large response at matching natural frequency.
- Periodic loudness variation from close frequencies.
- higher
- There is essentially no atmosphere to carry ordinary sound.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 14 - CORE PHYSICS

Fluid Statics and Dynamics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Calculate pressure in fluids.
- Apply buoyancy and Archimedes' principle.
- Use continuity and Bernoulli ideas for ideal flow.

Concept	Explanation	Formula / Example
Pressure	Normal force per unit area.	$P = F/A$
Hydrostatic pressure	Pressure increases with depth.	$\Delta P = \rho g h$
Buoyancy	Upthrust equals weight of displaced fluid.	$F_b = \rho_{\text{fluid}} g V_{\text{displaced}}$
Continuity	For incompressible steady flow, volume rate is conserved.	$A_1 v_1 = A_2 v_2$

Key Rules

- At the same depth in a connected static fluid, pressure is equal.
- An object floats when average density is low enough for buoyancy to balance weight.
- Bernoulli equation assumes steady, incompressible, nonviscous flow along a streamline.

Worked Example

Water density 1000 kg/m^3 . At depth 2.0 m , gauge pressure increase $= \rho g h = 1000(9.8)(2) = 19600 \text{ Pa}$.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 14 - PRACTICE

Fluid Statics and Dynamics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Pressure depends on container shape	At a given depth it depends on ρ , g , h	Hydrostatic pressure is local.
Buoyant force equals object weight always	Only for floating/equilibrium cases	In general it equals displaced fluid weight.

Guided Practice

- Write pressure formula.
- Find gauge pressure at 5 m in water.
- State Archimedes' principle.
- What determines whether an object floats?
- Write continuity equation.
- If pipe area halves, what happens to speed for incompressible steady flow?
- What is the SI unit of pressure?
- Name one limitation of ideal Bernoulli flow.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $P = F/A$
- 49000 Pa
- Buoyant force equals weight of displaced fluid.
- Density and displaced volume relative to fluid.
- $A_1 v_1 = A_2 v_2$
- It doubles.
- pascal (Pa)
- Viscosity/turbulence/compressibility can matter.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 15 - CORE PHYSICS

Thermal Physics and Calorimetry

Concepts, formulae, key rules, and worked example

Learning Objectives

- Distinguish temperature, heat, and internal energy.
- Calculate sensible and latent heat transfers.
- Apply thermal equilibrium and calorimetry.

Concept	Explanation	Formula / Example
Temperature	Measure related to thermal state and particle motion.	Kelvin is SI temperature unit
Specific heat	Energy needed per kg per kelvin temperature change.	$Q = mc \Delta T$
Latent heat	Energy absorbed/released during phase change at constant temperature.	$Q = mL$
Thermal equilibrium	No net heat flow once temperatures equalize.	zeroth-law idea

Key Rules

- Heat is energy in transit because of temperature difference.
- During an ideal phase change, added heat can change state without changing temperature.
- In an isolated calorimetry system, heat lost equals heat gained.

Worked Example

To heat 2.0 kg of water by 5 K with $c=4200 \text{ J/kg K}$: $Q=mc\Delta T=2(4200)(5)=42000 \text{ J}$.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

LESSON 15 - PRACTICE

Thermal Physics and Calorimetry

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Objects contain heat	Objects have internal energy; heat is transfer	Heat is process energy.
Celsius degree and Kelvin are unrelated	Temperature differences have same numerical size	1 degC interval = 1 K interval.

Guided Practice

- Write $Q = mc\Delta T$.
- Find heat for 1 kg water, $\Delta T = 10$ K, $c = 4200$.
- Write latent heat equation.
- What happens to temperature during ideal melting?
- Define thermal equilibrium.
- What direction does heat naturally flow?
- State conservation relation in calorimetry.
- SI unit of specific heat capacity?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $Q = mc\Delta T$
- 42000 J
- $Q = mL$
- It remains constant during the phase change.
- Equal temperature with no net heat transfer.
- hotter to colder
- heat lost = heat gained
- J/(kg K)

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 16 - CORE PHYSICS

Kinetic Theory and the Ideal Gas

Concepts, formulae, key rules, and worked example

Learning Objectives

- Use the ideal gas equation.
- Interpret gas pressure and temperature microscopically.
- Relate absolute temperature to average molecular kinetic energy.

Concept	Explanation	Formula / Example
Ideal gas law	Connects pressure, volume, amount, and absolute temperature.	$PV = nRT$
Microscopic pressure	Results from molecular collisions with container walls.	more/faster collisions $\rightarrow$ greater pressure
Absolute temperature	Kelvin temperature measures thermal scale from absolute zero.	$T(K) = T(C) + 273.15$
Molecular energy	Average translational kinetic energy is proportional to T .	$\langle K \rangle = \frac{3}{2} kT$

Key Rules

- Use kelvin, not Celsius, in gas-law ratios and $PV=nRT$.
- Ideal-gas assumptions work best at low density and away from condensation.
- At fixed volume and amount, pressure is proportional to absolute temperature.

Worked Example

For fixed n and V , if T rises from 300 K to 450 K, pressure scales by $450/300=1.5$, so it increases by 50%.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 16 - PRACTICE

Kinetic Theory and the Ideal Gas

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Use 20 in gas law for 20 degC	Use 293 K	Gas laws require absolute temperature.
Molecules stop at 0 degC	Absolute zero is 0 K	0 degC is 273.15 K.

Guided Practice

- Write ideal gas equation.
- Convert 27 degC to kelvin.
- At fixed V, P is proportional to what?
- At fixed P, V is proportional to what?
- What causes gas pressure microscopically?
- How does average molecular KE depend on T?
- Name one ideal-gas assumption.
- What happens to density if same mass expands to larger volume?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $PV=nRT$
- 300.15 K (about 300 K)
- absolute temperature
- absolute temperature
- molecular collisions with walls
- directly proportional
- Particles are point-like with negligible intermolecular forces except collisions.
- It decreases.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 17 - CORE PHYSICS

Laws of Thermodynamics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Apply the first law to thermal processes.
- Distinguish heat, work, and internal-energy changes.
- Explain the directionality expressed by the second law.

Concept	Explanation	Formula / Example
First law	Energy conservation for a thermodynamic system.	$\Delta U = Q - W_{\text{by}}$
Isothermal	Temperature remains constant.	For ideal gas, $\Delta U = 0$
Adiabatic	No heat transfer.	$Q = 0$
Second law	Natural processes have a preferred direction; heat engines cannot be 100% efficient.	entropy of isolated system does not decrease

Key Rules

- State your sign convention for work before calculations.
- For an ideal gas, internal energy depends on temperature.
- A heat engine must reject some heat to a colder reservoir.

Worked Example

A gas absorbs 500 J of heat and does 200 J of work on surroundings. $\Delta U = Q - W = 500 - 200 = 300 \text{ J}$.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 17 - PRACTICE

Laws of Thermodynamics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
First law says heat is conserved	Energy is conserved	Heat and work are transfer modes.
Engine efficiency can exceed 100%	Efficiency is less than 100%	Second law and energy conservation prohibit it.

Guided Practice

- Write first law using work done by system.
- $Q=800\text{ J}$, $W=300\text{ J}$. Find ΔU .
- What is Q in an adiabatic process?
- For ideal gas isothermal process, what is ΔU ?
- State one implication of second law.
- Why does a refrigerator require work?
- Define thermal efficiency conceptually.
- Can all absorbed heat be converted to work in a cyclic engine?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $\Delta U = Q - W$
- 500 J
- 0
- 0
- Entropy of an isolated system does not decrease / heat naturally flows hot to cold.
- It transfers heat from cold to hot, against spontaneous direction.
- useful work output / heat input
- No.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Electrostatics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Apply Coulomb's law for point charges.
- Calculate electric field and force.
- Use superposition for multiple charges.

Concept	Explanation	Formula / Example
Charge	Electric charge is conserved and quantized.	$q = ne$
Coulomb force	Force between point charges.	$F = k q_1q_2 /r^2$
Electric field	Force per unit positive test charge.	$E = F/q = kQ/r^2$
Superposition	Net force/field is vector sum of contributions.	$E_{net} = \Sigma E_i$

Key Rules

- Like charges repel and unlike charges attract.
- Electric field direction is the direction a positive test charge would accelerate.
- Use vector addition for fields and forces from more than one source.

Worked Example

A 2 microC test charge in a field of 3000 N/C experiences $F=qE=2\times10^{-6}(3000)=0.006$ N in the field direction.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

Electrostatics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Electric field depends on test charge	Field is set by source charges	Test charge only probes the field.
Add field magnitudes blindly	Add vectors	Directions matter.

Guided Practice

1. State Coulomb's law.

2. How does force change if distance doubles?

3. Define electric field.

4. Find force on 3 microC in 2000 N/C.

5. Direction of field around positive point charge?

6. Direction around negative charge?

7. State charge conservation.

8. What principle is used for multiple charges?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. $F=k|q_1q_2|/r^2$

2. one quarter

3. force per unit positive test charge

4. 0.006 N

5. outward

6. inward

7. Net electric charge of an isolated system remains constant.

8. superposition

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Electric Potential and Capacitance

Concepts, formulae, key rules, and worked example

Learning Objectives

- Relate electric potential, potential energy, and work.
- Calculate capacitance and stored energy.
- Analyze basic capacitor combinations.

Concept	Explanation	Formula / Example
Electric potential	Potential energy per unit charge.	$V = U/q$
Potential difference	Work per unit charge between points.	$\Delta V = \Delta U/q$
Capacitance	Charge stored per unit potential difference.	$C = Q/V$
Capacitor energy	Energy stored in electric field.	$U = 0.5CV^2$

Key Rules

- Electric potential is scalar; electric field is vector.
- For parallel plates, increasing plate area generally increases capacitance; increasing separation decreases it.
- Capacitors in parallel add directly; series combination has reciprocal addition.

Worked Example

A 10 microF capacitor at 12 V stores $Q=CV=10 \times 10^{-6}(12)=120 \text{ microC}$.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

Electric Potential and Capacitance

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Potential is force	Potential is energy per charge	Its unit is volt = J/C.
Capacitance depends on stored charge only	Geometry/material set C for a linear capacitor	Q changes with V while C stays fixed.

Guided Practice

1. Define electric potential.

2. What is 1 volt in base energy/charge terms?

3. Write $C=Q/V$.

4. Find Q for $C=5\text{ microF}$, $V=20\text{ V}$.

5. Write capacitor energy formula.

6. Two capacitors in parallel: how combine?

7. Two capacitors in series: how combine?

8. What is the unit of capacitance?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. potential energy per unit charge

2. 1 J/C

3. $C=Q/V$

4. 100 microC

5. $U=0.5CV^2$

6. $Ceq=C1+C2+...$

7. $1/Ceq=1/C1+1/C2+...$

8. farad (F)

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 20 - CORE PHYSICS

Direct-Current Circuits

Concepts, formulae, key rules, and worked example

Learning Objectives

- Use current, voltage, resistance, and Ohm's law.
- Combine resistors in series and parallel.
- Calculate electrical power and energy.

Concept	Explanation	Formula / Example
Current	Rate of flow of charge.	$I = \Delta Q / \Delta t$
Ohm's law	For an ohmic conductor at constant conditions.	$V = IR$
Series resistors	Same current; resistances add.	$R_{eq} = R_1 + R_2 + \dots$
Parallel resistors	Same voltage; reciprocal resistance adds.	$1/R_{eq} = 1/R_1 + 1/R_2 + \dots$

Key Rules

- Conventional current is defined in the direction positive charge would move.
- An ideal ammeter is connected in series; ideal voltmeter in parallel.
- Electrical power can be written $P = VI = I^2 R = V^2 / R$ for resistive elements.

Worked Example

A 6 ohm resistor across 12 V carries $I = V/R = 2$ A. Power = $VI = 24$ W.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

Direct-Current Circuits

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Voltmeter in series	Voltmeter in parallel	It measures potential difference across components.
Parallel resistance exceeds each branch	Equivalent is less than the smallest branch	More current paths reduce total resistance.

Guided Practice

1. Define electric current.

2. Find I for 10 V across 5 ohm.

3. Find series equivalent of 2, 3, 5 ohm.

4. Find parallel equivalent of two 6 ohm resistors.

5. Where connect ammeter?

6. Where connect voltmeter?

7. Find power at 12 V and 3 A.

8. State Ohm's law condition.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. rate of charge flow

2. 2 A

3. 10 ohm

4. 3 ohm

5. series

6. parallel

7. 36 W

8. V proportional to I at constant physical conditions for an ohmic conductor.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Kirchhoff's Laws and Electrical Power

Concepts, formulae, key rules, and worked example

Learning Objectives

- Apply junction and loop laws to multi-loop circuits.
- Use emf and internal resistance concepts.
- Calculate power delivery and dissipation.

Concept	Explanation	Formula / Example
Junction law	Charge conservation at a node.	sum currents in = sum currents out
Loop law	Energy conservation around a closed circuit.	sum Delta V = 0
EMF	Energy supplied per unit charge by a source.	emf in volts
Internal resistance	Real source terminal voltage drops under load.	$V_{\text{terminal}} = \text{emf} - Ir$

Key Rules

- Assign current directions arbitrarily; a negative solution means the true direction is opposite.
- Track voltage rises and drops consistently around each loop.
- Source power equals emf times current; delivered terminal power equals $V_{\text{terminal}} I$.

Worked Example

A 12 V battery with internal resistance 1 ohm drives a 5 ohm load. $I = 12 / (5 + 1) = 2$ A. Terminal voltage $= 12 - 2(1) = 10$ V.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

Kirchhoff's Laws and Electrical Power

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Current is consumed at a resistor	Charge flow is continuous	Resistors dissipate energy, not charge.
Loop voltage drops need not balance	Algebraic sum around loop is zero	Energy is conserved per unit charge.

Guided Practice

1. State Kirchhoff junction law.

2. State loop law.

3. What physical law supports junction rule?

4. What supports loop rule?

5. Find current for emf 9 V, r=1 ohm, R=2 ohm.

6. Find terminal voltage in that case.

7. What is source power formula?

8. If calculated current is negative, what does it mean?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. sum in = sum out

2. sum potential changes around loop = 0

3. charge conservation

4. energy conservation

5. 3 A

6. 6 V

7. P_source = emf I

8. Actual current direction is opposite to the assumed direction.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Magnetic Fields and Forces

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe magnetic fields produced by currents and magnets.
- Calculate force on moving charges and current-carrying wires.
- Analyze circular motion of charged particles in uniform fields.

Concept	Explanation	Formula / Example
Magnetic force on charge	Force is perpendicular to v and B.	$F = qvB \sin(\theta)$
Force on wire	Current in a magnetic field experiences force.	$F = BIL \sin(\theta)$
Circular path	Magnetic force can provide centripetal force.	$r = mv/(q B)$
Field direction	Use a right-hand rule for positive charge/current.	negative charge reverses direction

Key Rules

- A magnetic field does no work on a point charge when force is always perpendicular to velocity.
- Maximum magnetic force occurs when motion/current is perpendicular to B.
- Zero magnetic force occurs when motion/current is parallel to B.

Worked Example

A charge 2 microC moves at 1000 m/s perpendicular to B=0.5 T. $F=qvB=2 \times 10^{-6}(1000)(0.5)=0.001$ N.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

Magnetic Fields and Forces

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Magnetic force changes speed directly	It changes direction when perpendicular	Perpendicular force does no work.
Use left/right-hand rule without sign check	Reverse for negative charge	Rules usually give positive-charge direction.

Guided Practice

1. Write magnetic force on a moving charge.

2. When is it maximum?

3. When is it zero?

4. Write force on a straight current wire.

5. Can a static charge feel magnetic force from B alone?

6. What path for v perpendicular to uniform B?

7. Write radius formula.

8. Does magnetic force do work in ideal circular motion?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. $F=qvB \sin(\theta)$

2. $\theta=90^\circ$

3. $\theta=0^\circ$ or 180°

4. $F=BIL \sin(\theta)$

5. No, $v=0$ gives zero magnetic force.

6. circle

7. $r=mv/(|q|B)$

8. No.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Electromagnetic Induction

Concepts, formulae, key rules, and worked example

Learning Objectives

- Calculate magnetic flux.
- Apply Faraday's law and Lenz's law.
- Explain generators and motional emf.

Concept	Explanation	Formula / Example
Magnetic flux	Measures field passing through an area.	$\Phi = BA \cos(\theta)$
Faraday law	Changing flux induces emf.	$\text{emf} = -N \Delta \Phi / \Delta t$
Lenz law	Induced current opposes the change in flux that produces it.	minus sign in Faraday law
Motional emf	A conductor moving across B can develop emf.	$\text{emf} = Blv$ for perpendicular geometry

Key Rules

- Induction requires changing magnetic flux, not simply the presence of a magnetic field.
- Lenz's law is consistent with energy conservation.
- Flux angle is measured relative to the area normal in $\Phi = BA \cos(\theta)$.

Worked Example

A 100-turn coil has flux per turn change by 0.002 Wb in 0.10 s. Induced emf magnitude = $N \Delta \Phi / \Delta t = 100(0.002)/0.10 = 2 \text{ V}$.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 23 - PRACTICE

Electromagnetic Induction

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Constant flux induces steady emf	No emf if flux is constant	Faraday law depends on rate of change.
Lenz current aids the change	It opposes the change	Otherwise energy conservation would be violated.

Guided Practice

1. Define magnetic flux.

2. Write Faraday's law magnitude.

3. State Lenz's law.

4. What is unit of magnetic flux?

5. Can a stationary coil in constant B have zero induced emf?

6. How can flux be changed?

7. Write motional emf for perpendicular motion.

8. Why is Lenz law important physically?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. $\Phi = BA \cos(\theta)$

2. $|\text{emf}| = N|\Delta\Phi/\Delta t|$

3. Induced current opposes the flux change.

4. weber (Wb)

5. Yes.

6. Change B, area, angle, or move relative to field.

7. $\text{emf} = Blv$

8. It enforces energy conservation and gives current direction.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 24 - CORE PHYSICS

Alternating Current and Transformers

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe sinusoidal voltage and current.
- Use rms values for AC power calculations.
- Apply the ideal transformer ratio.

Concept	Explanation	Formula / Example
Alternating current	Current reverses direction periodically.	$i = I_0 \sin(\omega t)$
RMS value	DC-equivalent heating value for a sinusoid.	$V_{rms} = V_0 / \sqrt{2}$
Transformer	Uses changing magnetic flux to change AC voltage.	$V_s / V_p = N_s / N_p$
Ideal power	Input and output powers are equal ideally.	$V_p I_p = V_s I_s$

Key Rules

- A transformer requires changing flux and therefore operates with AC or varying current.
- Step-up voltage implies more secondary turns and lower secondary current ideally.
- Use rms values for ordinary AC power in resistive loads.

Worked Example

An ideal transformer has $N_p=200$, $N_s=1000$, $V_p=24$ V. $V_s=V_p(N_s/N_p)=24(5)=120$ V.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

LESSON 24 - PRACTICE

Alternating Current and Transformers

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Transformer works with steady DC	Steady DC gives no continuous induction	Flux must change.
Step-up voltage increases power ideally	Ideal power is unchanged	Current decreases correspondingly.

Guided Practice

- What does AC mean?
- Write V_{rms} for sinusoidal V_0 .
- Find V_{rms} if $V_0=100$ V.
- Write transformer voltage ratio.
- $N_p=500$, $N_s=100$, $V_p=230$ V. Find V_s .
- What happens to current in ideal step-up transformer?
- Why use laminated cores?
- Does ideal transformer conserve power?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- alternating current
- $V_0/\sqrt{2}$
- 70.7 V
- $V_s/V_p=N_s/N_p$
- 46 V
- It decreases.
- To reduce eddy-current losses.
- Yes, approximately.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Geometrical Optics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Apply reflection and refraction laws.
- Use Snell's law and refractive index.
- Analyze thin lenses and mirrors with ray diagrams and equations.

Concept	Explanation	Formula / Example
Reflection	Angle of incidence equals angle of reflection.	$\theta_i = \theta_r$
Refraction	Wave changes direction when speed changes across media.	$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$
Refractive index	Ratio of light speed in vacuum to speed in medium.	$n = c/v$
Thin lens	Relates focal length, object distance, image distance.	$1/f = 1/d_o + 1/d_i$

Key Rules

- Angles in reflection/refraction are measured from the normal, not the surface.
- Light bends toward the normal when entering a higher-index medium.
- Sign conventions matter in lens calculations; interpret image type after solving.

Worked Example

A converging lens has $f=10$ cm and object distance 30 cm. $1/d_i=1/10-1/30=2/30$, so $d_i=15$ cm: a real image.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 25 - PRACTICE

Geometrical Optics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Measure angle from surface	Measure from normal	Optics laws use the normal.
Higher n means faster light	Higher n means slower light	$n=c/v$.

Guided Practice

1. State law of reflection.
2. Write Snell's law.
3. Define refractive index.
4. What happens entering higher n at oblique incidence?
5. Write thin-lens equation.
6. What is focal point?
7. Can a converging lens form a virtual image?
8. What condition causes total internal reflection?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. $\theta_i = \theta_r$
2. $n_1 \sin \theta_1 = n_2 \sin \theta_2$
3. $n=c/v$
4. Bends toward normal.
5. $1/f = 1/d_o + 1/d_i$
6. Point where parallel rays converge or appear to diverge from.
7. Yes, if object is inside focal length.
8. From higher to lower n and incidence angle exceeds critical angle.

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

LESSON 26 - CORE PHYSICS

Wave Optics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Explain interference and diffraction.
- Apply double-slit fringe relations.
- Describe polarization as evidence for transverse waves.

Concept	Explanation	Formula / Example
Interference	Superposition produces constructive and destructive patterns.	path difference $m\lambda$ $\rightarrow$ constructive
Double slit	Fringe spacing for small angles.	$y = \lambda L/d$
Diffraction	Waves spread when passing apertures/edges, strongest when size $\sim$ wavelength.	single-slit minima: $a \sin \theta = m\lambda$
Polarization	Restriction of transverse oscillation direction.	supports transverse nature of light

Key Rules

- Stable interference needs coherent sources with fixed phase relationship.
- Longer wavelength or greater screen distance increases fringe spacing.
- Only transverse waves can be polarized in the usual sense.

Worked Example

For $\lambda=600$ nm, $L=2.0$ m, slit separation $d=0.50$ mm: $y=\lambda L/d = 2.4 \times 10^{-3}$ m = 2.4 mm.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 26 - PRACTICE

Wave Optics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Diffraction only occurs for light	All waves diffract	Extent depends on wavelength/aperture ratio.
Polarization proves light is longitudinal	It supports transverse behavior	Longitudinal waves cannot be plane-polarized similarly.

Guided Practice

1. Define constructive interference.
2. Define destructive interference.
3. Write double-slit fringe spacing.
4. How does y change if λ doubles?
5. How does y change if slit separation doubles?
6. When is diffraction strong?
7. What is coherence?
8. What property demonstrates transverse nature of light?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. Waves add in phase to larger amplitude.
2. Waves combine out of phase to reduce/cancel amplitude.
3. $y = \lambda L/d$
4. It doubles.
5. It halves.
6. When aperture/obstacle size is comparable to wavelength.
7. Fixed phase relationship and same frequency.
8. polarization

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Electromagnetic Waves

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe the electromagnetic spectrum.
- Relate wavelength, frequency, and speed of EM waves.
- Connect spectrum regions with applications and hazards.

Concept	Explanation	Formula / Example
EM wave	Oscillating electric and magnetic fields propagate through space.	$c = f \lambda$ in vacuum
Spectrum order	Radio, microwave, infrared, visible, ultraviolet, X-ray, gamma.	frequency increases across this order
Photon energy	Energy is proportional to frequency.	$E = hf$
Interactions	Different frequencies interact differently with matter.	ionizing risk rises at high photon energies

Key Rules

- All electromagnetic waves travel at c in vacuum.
- Higher frequency means shorter wavelength in the same medium.
- Ultraviolet, X-rays, and gamma rays can be ionizing depending on photon energy.

Worked Example

A radio wave of $f=100\text{ MHz}$ has $\lambda=c/f = 3.0\times10^8 / 1.0\times10^8 = 3.0\text{ m}$.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

LESSON 27 - PRACTICE

Electromagnetic Waves

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Radio waves travel slower than gamma in vacuum	All EM waves travel at c in vacuum	They differ in frequency and wavelength.
Higher frequency means longer wavelength	Higher frequency means shorter wavelength	$c = f\lambda$.

Guided Practice

- Write EM wave speed relation.
- List spectrum from low to high frequency.
- Find λ for 3 GHz in vacuum.
- Which has higher photon energy: red or violet?
- Do EM waves need a material medium?
- Name one microwave use.
- Name one X-ray use.
- Which constant links photon energy to frequency?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $c = f\lambda$
- radio, microwave, infrared, visible, ultraviolet, X-ray, gamma
- 0.10 m
- violet
- No.
- communications/radar/cooking
- medical imaging/security/material inspection
- Planck constant h

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Atomic Physics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe evidence for quantized atomic energy levels.
- Use photon energy relations.
- Explain emission and absorption spectra and basic photoelectric ideas.

Concept	Explanation	Formula / Example
Energy levels	Electrons in atoms occupy discrete allowed energies.	transitions emit/absorb photons
Photon	Quantum of electromagnetic radiation.	$E = hf = hc/\lambda$
Spectra	Line spectra reveal characteristic atomic energy differences.	$\Delta E = hf$
Photoelectric effect	Electrons are emitted if photon energy exceeds work function.	$K_{\max} = hf - \phi$

Key Rules

- Photon energy depends on frequency, not intensity.
- Increasing light intensity at fixed frequency increases photon rate, not energy per photon.
- Atomic line spectra support quantized energy levels.

Worked Example

A photon with $f=6.0 \times 10^{14}$ Hz has energy $E=hf = 6.63 \times 10^{-34} \times 6.0 \times 10^{14} = 3.98 \times 10^{-19}$ J.

Study Tip

Read the formula, identify every symbol and unit, then solve one new example without looking back.

Your Example / Notes

LESSON 28 - PRACTICE

Atomic Physics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Brighter light always ejects electrons	Frequency must exceed threshold	Photon energy must be sufficient.
Atoms emit any wavelength continuously	Isolated atoms give discrete lines	Energy levels are quantized.

Guided Practice

- Write photon energy formula.
- What does a line spectrum indicate?
- What is work function?
- State photoelectric threshold idea.
- Does intensity change energy per photon?
- What happens when an electron drops to a lower level?
- Write $K_{\max}$ relation.
- Which has more energy per photon: UV or infrared?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

- $E = hf = hc/\lambda$
- Discrete atomic energy levels.
- Minimum energy needed to remove an electron from a surface.
- Emission occurs only if $hf \geq \phi$.
- No.
- A photon is emitted with energy equal to the level difference.
- $K_{\max} = hf - \phi$
- UV

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Nuclear Physics

Concepts, formulae, key rules, and worked example

Learning Objectives

- Describe nuclear composition, isotopes, and radioactive decay.
- Use half-life and decay equations.
- Relate mass defect to binding energy.

Concept	Explanation	Formula / Example
Nucleus	Contains protons and neutrons; isotope differs in neutron number.	$A = Z + N$
Radioactive decay	Unstable nuclei transform spontaneously.	$N = N_0 e^{(-\lambda t)}$
Half-life	Time for number/activity to fall to half.	$T_{1/2} = \ln 2 / \lambda$
Mass-energy	Mass defect corresponds to binding energy.	$E = \Delta m c^2$

Key Rules

- Radioactive decay is random for a single nucleus but statistically predictable for large populations.
- Activity is proportional to number of undecayed nuclei.
- Conserve charge number and nucleon number in balanced nuclear equations.

Worked Example

A sample has half-life 4 h. After 12 h = 3 half-lives, remaining fraction = $(1/2)^3 = 1/8 = 12.5\%$.

Study Tip

Draw a diagram or free-body sketch before choosing equations.

Your Example / Notes

LESSON 29 - PRACTICE

Nuclear Physics

Common errors, guided practice, and mastery check

Common error	Better form	Why?
Half-life changes when sample amount changes	Half-life is a property of the isotope	Initial quantity changes activity, not half-life.
Radioactive decay can be predicted exactly for each nucleus	Individual decay time is random	Only statistical behavior is predictable.

Guided Practice

1. Define isotope.
2. What is half-life?
3. After 2 half-lives, what fraction remains?
4. After 4 half-lives?

5. Write relation between half-life and lambda.
6. What is activity?
7. Write mass-energy relation.
8. Name three common radiation types.

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. Same proton number, different neutron number.
2. Time for undecayed nuclei/activity to halve.
3. 1/4
4. 1/16

5. $T_{1/2} = \ln 2 / \lambda$
6. Decay rate, measured in becquerels.
7. $E = \Delta m c^2$
8. alpha, beta, gamma

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

Semiconductors and Modern Physics Review

Concepts, formulae, key rules, and worked example

Learning Objectives

- Explain intrinsic and doped semiconductor behavior.
- Describe diode and transistor roles qualitatively.
- Connect selected modern-physics ideas into a unified review.

Concept	Explanation	Formula / Example
Semiconductor	Conductivity lies between conductor and insulator and depends strongly on carriers.	silicon is common
Doping	Adding impurities creates n-type or p-type material.	n-type: electrons; p-type: holes
p-n junction	Diode allows strongly asymmetric current-voltage behavior.	forward bias conducts more readily
Transistor	Can act as an electronic switch or amplifier.	small control signal influences larger current

Key Rules

- A diode is not an ideal one-way wire; real devices have forward voltage and reverse limits.
- Semiconductor conductivity generally increases strongly with carrier concentration and temperature effects.
- Modern physics links quantization, waves, particles, atoms, nuclei, and electronic materials.

Worked Example

A simple LED circuit uses a series resistor to limit current. If supply is 5 V, LED drop is 2 V, and desired current is 0.015 A, $R=(5-2)/0.015 = 200\text{ ohm}$.

Study Tip

Check the final unit and estimate whether the magnitude is physically reasonable.

Your Example / Notes

Semiconductors and Modern Physics Review

Common errors, guided practice, and mastery check

Common error	Better form	Why?
LED can be connected directly to any voltage source	Use current limiting and correct polarity	Excess current can damage the device.
n-type material is negatively charged overall	Bulk material remains approximately neutral	Doping changes majority carriers, not net macroscopic charge.

Guided Practice

1. What is a semiconductor?

2. What are majority carriers in n-type material?

3. What are majority carriers in p-type?

4. What is a diode used for?

5. Why does an LED need current limiting?

6. Name one use of a transistor.

7. What physics idea explains atomic energy levels?

8. What relation connects mass and energy?

Mastery Challenge

- Solve two original numerical examples using the lesson principles.
- For each, write the governing equation before inserting numbers.
- Check units, significant figures, sign/direction, and physical reasonableness.

Quick Answers

1. Material with controllable intermediate conductivity.

2. electrons

3. holes

4. rectification/current-direction control

5. To prevent excessive current.

6. switching or amplification

7. quantization

8. $E=mc^2$

Exam Strategy

- Write known quantities with units before substituting numbers.
- Keep extra digits during working, then round the final answer appropriately.
- If a result has an impossible sign or unit, revisit the model, direction convention, and algebra.

Notes & Corrections

REVISION 01

Review Test 1 - Foundations & Kinematics

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Measurement, Units and Uncertainty	Convert 72 km/h to m/s.
2	Measurement, Units and Uncertainty	Write 0.0000450 m in scientific notation.
3	Measurement, Units and Uncertainty	How many significant figures are in 0.02030?
4	Scalars and Vectors	Classify distance: scalar or vector?
5	Scalars and Vectors	Classify acceleration.
6	Scalars and Vectors	Resolve 20 N at 60 degrees into x and y components.
7	Motion in One Dimension	Define instantaneous velocity.
8	Motion in One Dimension	A body moves from $x=2$ m to $x=-3$ m. Find displacement.
9	Motion in One Dimension	From rest, $a=2$ m/s ² for 5 s. Find v.
10	Projectile Motion	Resolve 30 m/s at 45 degrees into components.
11	Projectile Motion	What is vertical velocity at maximum height?
12	Projectile Motion	What is horizontal acceleration ideally?

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.
 Score 7-9: Good progress; review minor errors.
 Score 4-6: Revisit the related lessons.
 Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 02

Review Test 2 - Forces & Conservation

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Newton's Laws of Motion	State Newton's first law.
2	Newton's Laws of Motion	Find weight of 10 kg if $g=9.8 \text{ m/s}^2$.
3	Newton's Laws of Motion	A 4 kg body has net force 12 N. Find a.
4	Friction and Circular Motion	Write the formula for centripetal acceleration.
5	Friction and Circular Motion	A 2 kg mass moves at 4 m/s in radius 2 m. Find F_c .
6	Friction and Circular Motion	Direction of centripetal acceleration?
7	Gravitation	State Newton's law of gravitation.
8	Gravitation	How does force change if distance doubles?
9	Gravitation	Define gravitational field strength.
10	Work, Energy and Power	Find work by 10 N over 3 m in same direction.
11	Work, Energy and Power	Find kinetic energy of 4 kg at 5 m/s.
12	Work, Energy and Power	Find PE gain of 2 kg raised 3 m.

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 03

Review Test 3 - Momentum, Rotation & Oscillations

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Momentum and Collisions	Find momentum of 3 kg at 4 m/s.
2	Momentum and Collisions	A 10 N force acts 0.2 s. Find impulse.
3	Momentum and Collisions	What condition is needed for momentum conservation?
4	Rotational Motion and Torque	Define torque.
5	Rotational Motion and Torque	Find torque of 15 N perpendicular at 0.4 m.
6	Rotational Motion and Torque	What is the SI unit of torque?
7	Simple Harmonic Motion	State the SHM acceleration relation.
8	Simple Harmonic Motion	Where is speed maximum?
9	Simple Harmonic Motion	Where is acceleration magnitude maximum?
10	Mechanical Waves	Write the wave equation.
11	Mechanical Waves	Find speed for $f=20$ Hz, $\lambda=0.50$ m.
12	Mechanical Waves	Define wavelength.

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 04

Review Test 4 - Sound, Fluids & Thermal Physics

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Sound	Is sound transverse or longitudinal in air?
2	Sound	What determines pitch?
3	Sound	What determines intensity physically?
4	Fluid Statics and Dynamics	Write pressure formula.
5	Fluid Statics and Dynamics	Find gauge pressure at 5 m in water.
6	Fluid Statics and Dynamics	State Archimedes' principle.
7	Thermal Physics and Calorimetry	Write $Q = mc\Delta T$.
8	Thermal Physics and Calorimetry	Find heat for 1 kg water, $\Delta T = 10$ K, $c = 4200$.
9	Thermal Physics and Calorimetry	Write latent heat equation.
10	Kinetic Theory and the Ideal Gas	Write ideal gas equation.
11	Kinetic Theory and the Ideal Gas	Convert 27 degC to kelvin.
12	Kinetic Theory and the Ideal Gas	At fixed V, P is proportional to what?

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 05

Review Test 5 - Thermodynamics & Electrostatics

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Laws of Thermodynamics	Write first law using work done by system.
2	Laws of Thermodynamics	$Q=800\text{ J}$, $W=300\text{ J}$. Find ΔU .
3	Laws of Thermodynamics	What is Q in an adiabatic process?
4	Electrostatics	State Coulomb's law.
5	Electrostatics	How does force change if distance doubles?
6	Electrostatics	Define electric field.
7	Electric Potential and Capacitance	Define electric potential.
8	Electric Potential and Capacitance	What is 1 volt in base energy/charge terms?
9	Electric Potential and Capacitance	Write $C=Q/V$.
10	Direct-Current Circuits	Define electric current.
11	Direct-Current Circuits	Find I for 10 V across 5 ohm.
12	Direct-Current Circuits	Find series equivalent of 2, 3, 5 ohm.

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 06

Review Test 6 - Circuits, Magnetism & Induction

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Kirchhoff's Laws and Electrical Power	State Kirchhoff junction law.
2	Kirchhoff's Laws and Electrical Power	State loop law.
3	Kirchhoff's Laws and Electrical Power	What physical law supports junction rule?
4	Magnetic Fields and Forces	Write magnetic force on a moving charge.
5	Magnetic Fields and Forces	When is it maximum?
6	Magnetic Fields and Forces	When is it zero?
7	Electromagnetic Induction	Define magnetic flux.
8	Electromagnetic Induction	Write Faraday's law magnitude.
9	Electromagnetic Induction	State Lenz's law.
10	Alternating Current and Transformers	What does AC mean?
11	Alternating Current and Transformers	Write V_{rms} for sinusoidal V_0 .
12	Alternating Current and Transformers	Find V_{rms} if $V_0=100$ V.

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 07

Review Test 7 - Optics & Electromagnetic Waves

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Geometrical Optics	State law of reflection.
2	Geometrical Optics	Write Snell's law.
3	Geometrical Optics	Define refractive index.
4	Wave Optics	Define constructive interference.
5	Wave Optics	Define destructive interference.
6	Wave Optics	Write double-slit fringe spacing.
7	Electromagnetic Waves	Write EM wave speed relation.
8	Electromagnetic Waves	List spectrum from low to high frequency.
9	Electromagnetic Waves	Find λ for 3 GHz in vacuum.
10	Geometrical Optics	Write Snell's law.
11	Wave Optics	Write double-slit fringe spacing.
12	Electromagnetic Waves	Which has higher photon energy: red or violet?

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 08

Review Test 8 - Atomic, Nuclear & Semiconductor Physics

Mixed practice - complete without looking back at the lessons.

No.	Topic	Question
1	Atomic Physics	Write photon energy formula.
2	Atomic Physics	What does a line spectrum indicate?
3	Atomic Physics	What is work function?
4	Nuclear Physics	Define isotope.
5	Nuclear Physics	What is half-life?
6	Nuclear Physics	After 2 half-lives, what fraction remains?
7	Semiconductors and Modern Physics Review	What is a semiconductor?
8	Semiconductors and Modern Physics Review	What are majority carriers in n-type material?
9	Semiconductors and Modern Physics Review	What are majority carriers in p-type?
10	Atomic Physics	What does a line spectrum indicate?
11	Nuclear Physics	After 2 half-lives, what fraction remains?
12	Semiconductors and Modern Physics Review	What is a diode used for?

Writing / Explanation Task

Write an 80-120 word physics explanation using at least four ideas reviewed on this page. Include at least one equation and explain every symbol used.

Self-Assessment

Score 10-12: Strong mastery.

Score 7-9: Good progress; review minor errors.

Score 4-6: Revisit the related lessons.

Score 0-3: Study the lesson explanations before retrying.

Working Space

ANSWER KEY

Answer Key - Part 1

Use this section after completing the review tests independently.

Review Test 1 - Foundations & Kinematics

1. 20 m/s	7. Velocity at a particular instant.
2. 4.50×10^{-5} m	8. -5 m
3. 4	9. 10 m/s
4. scalar	10. 21.2 m/s, 21.2 m/s
5. vector	11. 0 m/s
6. 10 N, 17.3 N	12. 0 m/s^2

Review Test 2 - Forces & Conservation

1. Constant velocity unless acted on by a nonzero net force.	7. $F = Gm_1m_2/r^2$
2. 98 N	8. It becomes one quarter.
3. 3 m/s^2	9. Force per unit mass.
4. v^2/r	10. 30 J
5. 16 N	11. 50 J
6. toward the center	12. 58.8 J

Correction Method

- For every wrong answer, write the corrected result and the principle that explains it.
- If the same type of error repeats, return to the lesson and redo its Mastery Challenge.
- For numerical errors, identify whether the cause was model choice, algebra, units, signs, or rounding.

Error Log

ANSWER KEY

Answer Key - Part 2

Use this section after completing the review tests independently.

Review Test 3 - Momentum, Rotation & Oscillations

1. 12 kg m/s	7. $a = -\omega^2 x$
2. 2 N s	8. equilibrium
3. Negligible external impulse.	9. at maximum displacement
4. Turning effect of a force about an axis.	10. $v = f \lambda$
5. 6 N m	11. 10 m/s
6. N m	12. Distance between points in the same phase on successive cycles.

Review Test 4 - Sound, Fluids & Thermal Physics

1. longitudinal	7. $Q = mc\Delta T$
2. frequency	8. 42000 J
3. power per unit area / amplitude-related energy	9. $Q = mL$
4. $P = F/A$	10. $PV = nRT$
5. 49000 Pa	11. 300.15 K (about 300 K)
6. Buoyant force equals weight of displaced fluid.	12. absolute temperature

Correction Method

- For every wrong answer, write the corrected result and the principle that explains it.
- If the same type of error repeats, return to the lesson and redo its Mastery Challenge.
- For numerical errors, identify whether the cause was model choice, algebra, units, signs, or rounding.

Error Log

ANSWER KEY

Answer Key - Part 3

Use this section after completing the review tests independently.

Review Test 5 - Thermodynamics & Electrostatics

1. $\Delta U = Q - W$	7. potential energy per unit charge
2. 500 J	8. 1 J/C
3. 0	9. $C = Q/V$
4. $F = k q_1q_2 /r^2$	10. rate of charge flow
5. one quarter	11. 2 A
6. force per unit positive test charge	12. 10 ohm

Correction Method

- For every wrong answer, write the corrected result and the principle that explains it.
- If the same type of error repeats, return to the lesson and redo its Mastery Challenge.
- For numerical errors, identify whether the cause was model choice, algebra, units, signs, or rounding.

Error Log

ANSWER KEY

Answer Key - Part 4

Use this section after completing the review tests independently.

Review Test 6 - Circuits, Magnetism & Induction

1. sum in = sum out	7. $\Phi = BA \cos(\theta)$
2. sum potential changes around loop = 0	8. $ \mathcal{E} = N \left \frac{\Delta \Phi}{\Delta t} \right $
3. charge conservation	9. Induced current opposes the flux change.
4. $F = qvB \sin(\theta)$	10. alternating current
5. $\theta = 90^\circ$	11. $V_0 / \sqrt{2}$
6. $\theta = 0^\circ$ or 180°	12. 70.7 V

Correction Method

- For every wrong answer, write the corrected result and the principle that explains it.
- If the same type of error repeats, return to the lesson and redo its Mastery Challenge.
- For numerical errors, identify whether the cause was model choice, algebra, units, signs, or rounding.

Error Log

ANSWER KEY

Answer Key - Part 5

Use this section after completing the review tests independently.

Review Test 7 - Optics & Electromagnetic Waves

1. $\theta_i = \theta_r$	7. $c = f \lambda$
2. $n_1 \sin \theta_1 = n_2 \sin \theta_2$	8. radio, microwave, infrared, visible, ultraviolet, X-ray, gamma
3. $n = c/v$	9. 0.10 m
4. Waves add in phase to larger amplitude.	10. $n_1 \sin \theta_1 = n_2 \sin \theta_2$
5. Waves combine out of phase to reduce/cancel amplitude.	11. $y = \lambda L/d$
6. $y = \lambda L/d$	12. violet

Correction Method

- For every wrong answer, write the corrected result and the principle that explains it.
- If the same type of error repeats, return to the lesson and redo its Mastery Challenge.
- For numerical errors, identify whether the cause was model choice, algebra, units, signs, or rounding.

Error Log

ANSWER KEY

Answer Key - Part 6

Use this section after completing the review tests independently.

Review Test 8 - Atomic, Nuclear & Semiconductor Physics

1. $E=hf=hc/\lambda$	7. Material with controllable intermediate conductivity.
2. Discrete atomic energy levels.	8. electrons
3. Minimum energy needed to remove an electron from a surface.	9. holes
4. Same proton number, different neutron number.	10. Discrete atomic energy levels.
5. Time for undecayed nuclei/activity to halve.	11. $1/4$
6. $1/4$	12. rectification/current-direction control

Correction Method

- For every wrong answer, write the corrected result and the principle that explains it.
- If the same type of error repeats, return to the lesson and redo its Mastery Challenge.
- For numerical errors, identify whether the cause was model choice, algebra, units, signs, or rounding.

Error Log

REFERENCE

Formulae, Constants and SI Units

Core equations for rapid review. Always check the assumptions behind each formula.

Mechanics

$v = u + at$	$s = ut + 0.5at^2$
$v^2 = u^2 + 2as$	$\Sigma F = ma$
$W = Fd \cos(\theta)$	$K = 0.5mv^2$
$p = mv$	$J = \Delta p$
$F_c = mv^2/r$	$F_g = Gm_1m_2/r^2$

Oscillations, Waves & Thermal

$T_{\text{spring}} = 2\pi \sqrt{m/k}$	$v = f \lambda$
$Q = mc \Delta T$	$Q = mL$
$PV = nRT$	$\Delta U = Q - W_{\text{by}}$

Electricity & Magnetism

$F = k q_1q_2 /r^2$	$E = F/q$
$V = U/q$	$C = Q/V$
$U_{\text{cap}} = 0.5CV^2$	$V = IR$
$P = VI$	$F_B = qvB \sin(\theta)$
$\Phi = BA \cos(\theta)$	$\text{emf} = -N \Delta \Phi / \Delta t$

Optics & Modern Physics

$n = c/v$	$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$
$1/f = 1/d_o + 1/d_i$	$y = \lambda L/d$
$E_{\text{photon}} = hf = hc/\lambda$	$K_{\text{max}} = hf - \phi$
$N = N_0 e^{-(\lambda t)}$	$T_{1/2} = \ln 2/\lambda$
$E = mc^2$	

Selected Constants

Quantity	Symbol	Approximate value
Speed of light	c	3.00 x 10 ⁸ m/s
Gravitational constant	G	6.67 x 10 ⁻¹¹ N m ² /kg ²
Elementary charge	e	1.60 x 10 ⁻¹⁹ C
Planck constant	h	6.63 x 10 ⁻³⁴ J s
Gas constant	R	8.31 J/(mol K)
Earth surface gravity	g	9.81 m/s ²

REFERENCE

Physics Quick Guide

A one-page checklist for formulas, graphs, experiments, and exams.

Before Solving

- Sketch the situation and choose axes/directions.
- List known and unknown quantities with SI units.
- State the physical law/model and its assumptions.

During Calculation

- Rearrange symbolically before inserting numbers when practical.
- Carry units through the calculation.
- Keep enough digits during working; round at the end.

Graphs

- Read axis labels and units first.
- Slope and area have physical meanings only when derived from the plotted quantities.
- Distinguish a straight-line relationship from direct proportionality through the origin.

Experiments

- Identify independent, dependent, and controlled variables.
- Repeat measurements to assess random uncertainty.
- Separate random uncertainty from systematic bias and state improvements.

Explanation Questions

- Use cause-and-effect language: law $\rightarrow$ mechanism $\rightarrow$ observed result.
- Name directions and interactions explicitly.
- State limitations of ideal models such as no air resistance, point particles, or negligible losses.

Final Check

- Does the unit match the requested quantity?
- Is the sign/direction physically sensible?
- Is the magnitude reasonable compared with the data and limiting cases?